CE152: Object-Oriented Programming

Unit 4

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CSEE

2026

Labs

Marked lab exercise 2

- Exercise sheet is on Moodle in the **Lab Exercises (Week 19-21)** section
- The demonstrations are in the labs in **Week 20** or **Week 21** (next week and the week after)
- Demos are not possible this week
- Please use the time in Week 19 for the unmarked labs and for questions

Feedback on Marked lab exercise 1

- Around 200 of you demonstrated the exercise in the lab
- Thank you for all the good work and I hope the demonstrations and the feedback you got were/will be helpful for you!
- You should always attempt the exercise and the key is to prepare before the lab
- If you feel you did not have the information you needed please let me know how I could have provided it
- If factors outside of your control are preventing you from working on the exercises please notify the School Office

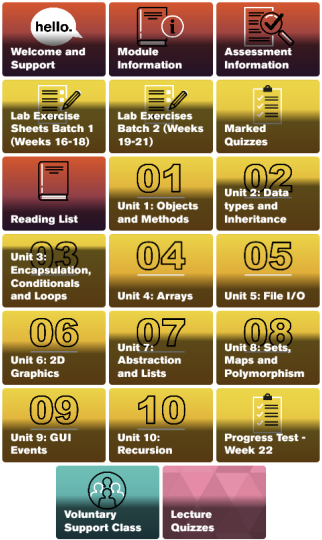
Completing the marked exercises

- Marked Exercise 2 is much more difficult and requires more work
- If it were easy society would not respect the skills you acquire as much (which will e.g. be reflected in the salary of your future job)
- Also keep in mind that for many second and third year modules you will require Java programming skills
- Thus your goal should be to learn Java programming and not just to pass this module somehow
- Otherwise the upcoming modules will be much, much harder because they will be focussing on difficult content going beyond the programming skills
- **In summary: attempting the marked exercises is one of the best ways to learn Java and also to get marks that contribute to passing the module!**

Suggestion for strategy for next Marked Lab Exercise

- First thing after this lecture download the PDFs
- Read the brief and identify the key tasks you need to be able to do, e.g.:
  - Creating and using classes, variables, constructors and methods (Units 1-3)
  - Creating and using arrays (Unit 4)
  - Reading and writing files (Unit 5)
- **Start preparing for this now (Week 19)**
- By Week 20 you should be ready to start practicing for the demonstration
- Demonstrate in Week 20 or in Week 21

Location of Lab Exercise Sheets Batch 2 on Moodle



Progress Test

Practice

- Remember that you need to know the content both from the pre-recorded videos and the Zoom lectures
- We will practice some questions next week and the week after
- The test is scheduled for Week 22 (during the labs)

Access modifiers

Table of access modifiers

Modifier	Class	Package	Subclass	World
`public`	Y	Y	Y	Y
`protected`	Y	Y	Y	N
no modifier	Y	Y	N	N
`private`	Y	N	N	N

Videos

Unit 4

- Please remember to watch the videos on **Arrays Part 1** and **Arrays Part 2**
- You can find these videos on Moodle in **Unit 4: Arrays**
- The lecture that is taking place now is not identical to the videos
- Remember there is a Quiz for Unit 4 in the marked quizzes section that can only be completed this week!

Quiz 3

Question 3 (1)

Consider the class `ConditionalPractice`.

```
public class ConditionalPractice {

}
```

Question 3 (2)

Add a method called `checkSign` that receives one `int` as parameter... a

```
public class ConditionalPractice {
    public int checkSign (int x) {

    }
}
```

Question 3 (3)

... and returns -1 if the `int` is negative, 0 if it is 0 and 1 if it is positive.

```
public class ConditionalPractice {
    public int checkSign (int x) {
        if (x<0)
            return -1;
        else if (x>0)
            return 1;
        else
            return 0;
    }
}
```

Question 4 (1)

Consider the class called `ConditionalPractice`.

```
public class ConditionalPractice {

}
```

Question 4 (2)

Add a method called `stringComparison` that receives two `String` as parameters and returns a `boolean` ...

```
public class ConditionalPractice {
    public boolean stringComparison(String a, String b) {

    }
}
```

Question 4 (3)

... that is true if the two strings have identical content and false if they do not.

```
public class ConditionalPractice {
    public boolean stringComparison(String a, String b) {
        return a.equals(b);
    }
}
```

Question 7 (1)

Write a `public` class called `Loops` . . .

```
public class Loops {

}
```

Question 7 (2)

... with a method called `printIntFrom10` that returns `void` and ...

```
public class Loops {
    public void printIntFrom10(int x) {

    }
}
```

Question 7 (3)

... prints all `int` larger or equal to 10 up until and including the number you supply as a parameter. The loop should print nothing if your number is smaller than 10.

```
public class Loops {
    public void printIntFrom10(int x) {
        for (int i=10; i<=x; i++) {
            System.out.println(i);
        }
    }
}
```

Question 10 (1)

Write a public class with a name of your choosing that has a `main` method...

```
public class MyScannerTest {
    public static void main(String[] args){

    }
}
```

Question 10 (2)

... in this method ask the user for input using Scanner and ...

```
import java.util.Scanner;

public class MyScannerTest {
    public static void main(String[] args){
        Scanner input =
            new Scanner(System.in).useDelimiter("\n");
        String value = input.next();
        int numIterations = Integer.parseInt(value);
    }
}
```

Question 10 (3)

... print the numbers from 0 up to and including that number to the command line. If the number is negative print nothing.

```
import java.util.Scanner;
public class MyScannerTest {
    public static void main(String[] args){
        Scanner input =
            new Scanner(System.in).useDelimiter("\n");
        String value = input.next();
        int numIterations = Integer.parseInt(value);
        for(int i=0; i<=numIterations; i++)
            System.out.println(i);
    }
}
```

Thank you!